

Jonathan Zhao

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EDUCATION

Massachusetts Institute of Technology (MIT)

Bachelor's Degree in Computer Science and Engineering

June 2026

Cambridge, MA

- GPA: 3.8; Relevant Coursework: Programming in C and Assembly, Design and Analysis of Algorithms, Software Design, Operating Systems Engineering, Natural Language Processing, Database Systems, Computer Systems Engineering

WORK EXPERIENCE

Simons Electron Microscopy Center

Software Engineering Intern

May 2025 – August 2025

New York, NY

- Designed Cryo-Insight, a Python library enabling Leginon users to convert cryo-EM datasets into EMinsight schema format for downstream analysis workflows
- Extended EMinsight with additional data aggregation, reporting, and visualization features

Duet

Backend Engineer

Nov 2024 – Feb 2025

Remote

- Developed a MusicXML parser and musical signature indicator identifier in Python using Music21, serving as a core data pipeline for an AI-based sheet music writing copilot

MIT Media Lab

Undergraduate Researcher, MIT Game Lab

Oct 2022 – June 2023

Cambridge, MA

- Designed player-facing ARG puzzles teaching DEI principles to high school students, blending physical puzzle mechanics with virtual presentation

Feil Family Brain & Mind Research Institute

Researcher, Gang Wang Lab

April 2019 – November 2021

New York, NY

- Researched voltage-gated L-type calcium channels in neuropsychiatric disorders in early-stage Alzheimer's Disease; co-discovered role of L-type channels and amygdala NPY neurons in anxiety-related behaviors

PROJECTS

GoDB — Relational Database Engine in Go

Spring 2026

- Built a relational database engine in Go, implementing heap file storage, buffer pool management, and a full operator suite (filter, join, aggregation with group-by, insert/delete)
- Implemented strict 2PL transactions with page-level lock management and deadlock detection; built a Selinger-style cost-based query optimizer using histogram-based selectivity estimation for join ordering

Why Transformers Miss Toxicity — NLP Interpretability Research

Spring 2025

- Fine-tuned BERT on 300K+ examples across 3 toxicity corpora; applied sparse autoencoders (SAEs) to decompose residual stream activations into 256 interpretable features, ranking toxic correlates via Pearson analysis and causal ablation; used TransformerLens to identify failure modes via attention head ablation and activation patching

WaitLess — Real-Time Nightlife Crowdsourcing Platform

Fall 2025

- Built full-stack venue crowdsourcing platform (React, Node.js/Express, MongoDB) with Socket.io real-time updates, configurable alert subscriptions triggered by user-defined wait time and crowd conditions
- Implemented geospatial indexing for location-aware queries and background snapshots aggregating crowd stats for peak time forecasting

OTHER EXPERIENCE

Sigma Alpha Epsilon Fraternity, Mass Iota-Tau Chapter

President

Jan 2025 – Jan 2026

Cambridge, MA

- Managed and led a 30+ member fraternity chapter; oversaw a \$100,000 annual budget for event planning and chapter operations

SKILLS & INTERESTS

- Technical: MERN stack (JS/TS, Node.js, MongoDB, Express), React, Socket.io, Python (pandas, NumPy, SQLAlchemy), ML Engineering (PyTorch, scikit-learn, HuggingFace), C, Assembly, AWS, SQL, Agile
- Languages: English (Fluent), Mandarin Chinese (Fluent), French (Basic)
- Interests: Poker, Boxing, Game Design, Film, Philosophy, Motorsports